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Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection

Study participant details. Participant data were obtained from four prospective clinical trials in RCC: JAVELIN Renal 101 (ClinicalTrials.gov identifier: NCT02684006), GETUG-AFU 26 NIVOREN (ClinicalTrials.gov identifier: NCT03013335), SURF (ClinicalTrials.gov identifier: NCT02689167), and ONCOBIOTICS (ClinicalTrials.gov identifier: NCT04567446).

Clinical trials
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disease progression, unacceptable toxicity, withdrawal of consent, or death. The primary endpoint was to assess the incidence of high-grade adverse events (Grades 3-4 and Grade 5, as per CTCAE v4.0) 4. Secondary endpoints included ORR, PFS, OS, and exploration of candidate blood biomarkers associated with response and/or prognosis to treatment. Blood samples were collected before and on different visits after Nivolumab start. Available baseline, cycle 3, cycle 7 and progression visit samples were used in this analysis.
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The reported clinical trials comply with all relevant ethical regulations on Research Involving Human Subjects.
Ethics approval and consent to participate. The study design and conduct complied with all relevant regulations regarding the use of human study participants and were conducted was conducted in accordance with the Declaration of Helsinki and Good Clinical Practice (GCP) guidelines. All participants provided written informed consent. Data was reported disaggregated for sex and gender. Consent has been obtained for reporting and sharing individual-level data. Data and sample collection adhered to regulatory, ethical requirements and ICH E6(R2) GCP guidelines.
Stool samples collection. Stool samples were self-collected by the participants at home according to the IHMS_SOP 004 (https://human-microbiome.org/index.php?id=Sop&num=004) within the ONCOBIOTICS, AtezoTRIBE and French cohort of the STRONG phase IIIb trial study protocols. Immediately after the collection, they were stored at −20 °C and transported on dry ice. Upon arriving at the biobank of the recruitment center (between 4 and 24 h later), they were stored at −80°C in plastic tubes. Two aliquots of the same sample were used for metaproteomic and metagenomic analyses. All the samples were processed according to International Human Microbiome Standards (IHMS) guidelines (SOP 03 V1).

Data analysis

Statistical methods. Quantitative data are described by the number of patients with available data, the number of missing values, the mean, standard deviation, median, minimum and maximum values, and the 1st and 3rd quartile values (Q1-Q3). sMAdCAM-1 values were compared between groups using the Mann–Whitney or Kruskal-Wallis test. Spearman rank correlation analyses between MAdCAM-1 and each of interleukin-6 (IL-6) and interleukin-8 (IL-8) were performed to evaluate the relationship between MAdCAM-1 and these inflammatory cytokines. Categorical groups (sMAdCAM-1 high vs low, or the dynamic groups from Cycle 1 to Cycle 3) were compared between treatments using the Chi-square test. The dynamic values from Cycle 1 to Cycle 3 (Fig. 4A) were compared Wilcoxon signed-rank test. OS was defined as the time, expressed in months, from treatment initiation to the date of death or the last known follow-up. Patients alive at the last follow-up date were censored. PFS was defined as the time, expressed in months, from treatment initiation to the date of the first event (progression or death). Patients were censored at last disease/tumor assessment. The optimal cutoff value was determined by the Contal and O'Quigley's approach for the OS outcome based on sMAdCAM-1 values at Cycle 1 in the training cohort 16. The assumption of linearity was visually checked by the Martingale residual plots as well as Restricted Cubic Spline plots (with 5 knots). The discrimination of the fitted model was assessed by time-dependent AUC at 18 months and Integrated AUC index across all follow up times (ranging 0-1, with higher numbers indicating better discrimination performance) 17. The 18-month timepoint was chosen based on the median follow-up of 18.9 months in this cohort. Kaplan-Meier curves and Cox proportional hazards models were used to compare PFS and OS across the sMAdCAM groups (high vs low at Cycle 1, or the dynamic groups from Cycle 1 to Cycle 3). The analysis at Cycle 3 was based on a landmark approach calculated from Cycle 3 (3-months) for OS and PFS. Analyses were performed using SAS 9.4 (SAS Institute Inc. Cary, NC), R packages ("rms" for Restricted Cubic Spline plot; "timeROC" for time-dependent ROC), and SAS Macro for determining the optimal cutoff (https://support.sas.com/resources/papers/proceedings/proceedings/sugi28/261-28.pdf).
Metagenomics analysis. DNA was extracted from an aliquot of the same stool samples used for metaproteomic analysis using the IHMS SOP 07 V2 H and sequenced with an Ion proton sequencer (Thermo Fisher Scientific). The sequencing was performed at a single site (MetaGenoPolis). Reads were cleaned using Alien Trimmer (version 0.4.0) to (1) remove resilient sequencing adapters and (2) trim low-quality nucleotides at the 3' side using a quality cutoff of 20 and a length cutoff of 45 bp. Cleaned reads were subsequently filtered from human and other possible food contaminant DNA using human genome (GRCh38-p13), with an identity score threshold of 95%. For each metagenome we profiled the taxonomic composition with MetaPhlAn-4 18. For alpha diversity, we computed the per-sample Richness and Shannon indexes 19 using the vegan R package 20. Differences between groups were assessed using the Wilcoxon-Mann-Whitney test. For beta-diversity, we computed the between-samples Bray-Curtis dissimilarities using the implementation available in the Vegan R package 20. Differential abundance analysis on taxonomic profiles between groups were performed using the LefSE software. The longitudinal analysis was performed with a Wilcoxon signed-rank test (testing only features present in at least 10 samples in one of the groups). None of the associations

presented adjusted p-values with Benjamini-Hochberg correction lower than 0.2. In figures, microbial features have been subset as the ones presenting 50% prevalence in both tested groups and raw p-value < 0.05.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

MetaPhlAn4 profiles and Luminex data for the cohorts included in this study are available via Figshare. The remaining data will be available within the Article, Supplementary Information or Source Data file.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Sex and/or gender were not considered in the studies design. Sex-at-birth (assigned) was collected per protocol detailed in previous publications. The sex, number and age of participants are described in Table 1.
Reporting on race, ethnicity, or other socially relevant groupings	Not reported.
Population characteristics	Relevant patient characteristics are described in Table 1 and previous published articles.
Recruitment	Participant data were obtained from four prospective clinical trials in RCC: JAVELIN Renal 101 (NCT02684006), GETUG-AFU 26 NIVOREN (NCT03013335), SURF (NCT02689167), and ONCOBIOTICS (NCT04567446). The study design and sample size have been reported peviously. In this paper, all patients with available stool and serum samples submitted to translational research were included.
Ethics oversight	Ethics approval and consent to participate. The study design and conduct complied with all relevant regulations regarding the use of human study participants and were conducted was conducted in accordance with the Declaration of Helsinki and Good Clinical Practice (GCP) guidelines. All participants provided written informed consent. Consent to publish clinical information potentially identifying individuals was obtained (age and sex). Data and sample collection adhered to regulatory, ethical requirements and ICH E6(R2) GCP guidelines.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Participant data were obtained from four prospective clinical trials in RCC: JAVELIN Renal 101 (NCT02684006), GETUG-AFU 26 NIVOREN (NCT03013335), SURF (NCT02689167), and ONCOBIOTICS (NCT04567446). The study design and sample size have been reported peviously. In this paper, all patients with available stool and serum samples collected for translational research purposes were included.
Data exclusions	We only excluded samples that dit not pass quality control check.
Replication	Direct replication via a separate cohort was not feasible due to the limited availability of samples from this specific clinical trial population. However, the robustness of the Luminex platform is well-established, and the findings are internally validated through their correlation with clinical outcomes.
Randomization	We confirm that for the biomarker analyses, group allocation was not random but was defined by data-driven biomarker cut-offs (e.g., high

Randomization	vs. low sMAdCAM-1). To account for potential confounding factors arising from this non-randomized allocation, we performed multivariable analyses adjusting for key clinical covariates such as IMDC risk group, age, and antibiotics use.
Blinding	Participant data were obtained from four prospective clinical trials in RCC: JAVELIN Renal 101 (NCT02684006), GETUG-AFU 26 NIVOREN (NCT03013335), SURF (NCT02689167), and ONCOBIOTICS (NCT04567446). We confirm that for the biomarker analyses, blinding was not relevant as the experimental groups were defined by data-driven biomarker cut-offs (e.g., high vs. low sMAdCAM-1). The group allocation itself was determined by the outcome of the measurement, making blinding of the investigators during data collection and analysis not applicable to this specific study design.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	Participant data were obtained from four prospective clinical trials in RCC: JAVELIN Renal 101 (ClinicalTrials.gov identifier: NCT02684006), GETUG-AFU 26 NIVOREN (ClinicalTrials.gov identifier: NCT03013335), SURF (ClinicalTrials.gov identifier: NCT02689167), and ONCOBIOTICS (ClinicalTrials.gov identifier: NCT04567446).
Study protocol	Clinical trials JAVELIN Renal 101 is a multicenter, randomized, open-label, phase III clinical trial evaluating the efficacy and safety of avelumab plus axitinib versus sunitinib monotherapy in patients with previously untreated advanced RCC with a clear-cell component. A total of 886 adults were enrolled and randomized 1:1 to receive either avelumab (10 mg/kg IV every 2 weeks) plus axitinib (5 mg orally twice daily) or sunitinib (50 mg orally once daily for 4 weeks in a 6-week cycle). For alignment within the study, a single cycle consisted of 6 weeks for both treatment arms. The trial design has been previously described, and results are reported per CONSORT guidelines 1,2. The co-primary endpoints were progression-free survival (PFS) and overall survival (OS) in patients with PD-L1–positive tumors 1,3. A key secondary endpoint was PFS in the overall population; other endpoints included objective response rate (ORR). Efficacy data reported here reflect the second interim analysis of the JAVELIN Renal 101 trial, with a data cutoff of January 2019. Blood samples were collected at screening and on multiple days within cycles 1, 2, and 3, prior to treatment. Available baseline and cycle 3 samples were used in this analysis. Collection included whole blood in dipotassium ethylenediaminetetraacetic acid (K_2EDTA) and silica-coated tubes for plasma and serum separation, respectively. Sample availability varied based on patient consent, local regulations, sample pairing, volume constraints, and site-level data reporting. The protocol has been previously published 1. GETUG-AFU 26 NIVOREN trial is a French, multicenter, prospective, phase II clinical trial evaluating the activity and safety of nivolumab in patients with metastatic ccRCC who had previously failed on or after an anti-angiogenic regimen, as previously published 4–6. Participants received nivolumab at 3 mg/kg every two weeks and underwent radiological assessment every 8–12 weeks. Treatment was continued until disease progression, unacceptable toxicity, withdrawal of consent, or death. The primary endpoint was to assess the incidence of high-grade adverse events (Grades 3-4 and Grade 5, as per CTCAE v4.0) 4. Secondary endpoints included ORR, PFS, OS, and exploration of candidate blood biomarkers associated with response and/or prognosis to treatment. Blood samples were collected before and on different visits after Nivolumab start. Available baseline, cycle 3, cycle 7 and progression visit samples were used in this analysis. SURF trial is a French prospective, randomized, open-label phase IIb study that included patients with metastatic ccRCC treated with sunitinib at a standard 50 mg daily for 4 weeks, followed by a 2-weeks rest period (4/2) 7. Patients requiring a dose adjustment due to toxicity were randomized between the standard 4/2 schedule at a reduced dose of 37.5 mg daily and the experimental 2/1 schedule at 50 mg daily. The primary objective was to assess the median duration of sunitinib treatment in each group. Key secondary objectives included PFS, OS, time to randomization, ORR, safety, sunitinib dose intensity, health-related quality of life, and the description of main drivers triggering randomization. An exploratory endpoint included the identification of blood biomarkers related to the activity and/or toxicity of sunitinib. Available baseline blood samples were used in this report. The protocol has been previously published 8. RCC ONCOBIOTICS is a multicentric study evaluating gut microbiota related biomarkers associated with outcome in advanced RCC treated with ICI alone or in combination with TKI at Gustave Roussy, in France. The study followed standard care until disease progression, unacceptable toxicity, or completion of 2 years of ICI treatment. Eligibility criteria and baseline data, including recent medications, are in the trial protocol and recorded in electronic case report forms. Blood and stool samples were collected before

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Outcomes

In the current work, we asked whether sMAdCAM-1 levels could serve as a biomarker of an immunosuppressive microbiota linked to resistance to ICI and/or TKI regimens specifically in patients with metastatic RCC (mRCC). We investigated the distribution and biological significance of baseline and on-treatment sMAdCAM-1 levels in three independent cohorts of patients with mRCC: the JAVELIN Renal 101 trial (NCT02684006) the SURF trial (NCT02689167) and the GETUG-AFU26-NIVOREN trial (NCT03013335). Overall Survival (OS) was defined as the total length of time from the start of treatment until death from any cause (months). Progression-Free Survival (PFS) was defined as the time from the start of treatment until the cancer progresses or the patient dies from any cause, whichever comes first (months).

Plants

Seed stocks

NA

Novel plant genotypes

NA

Authentication

NA